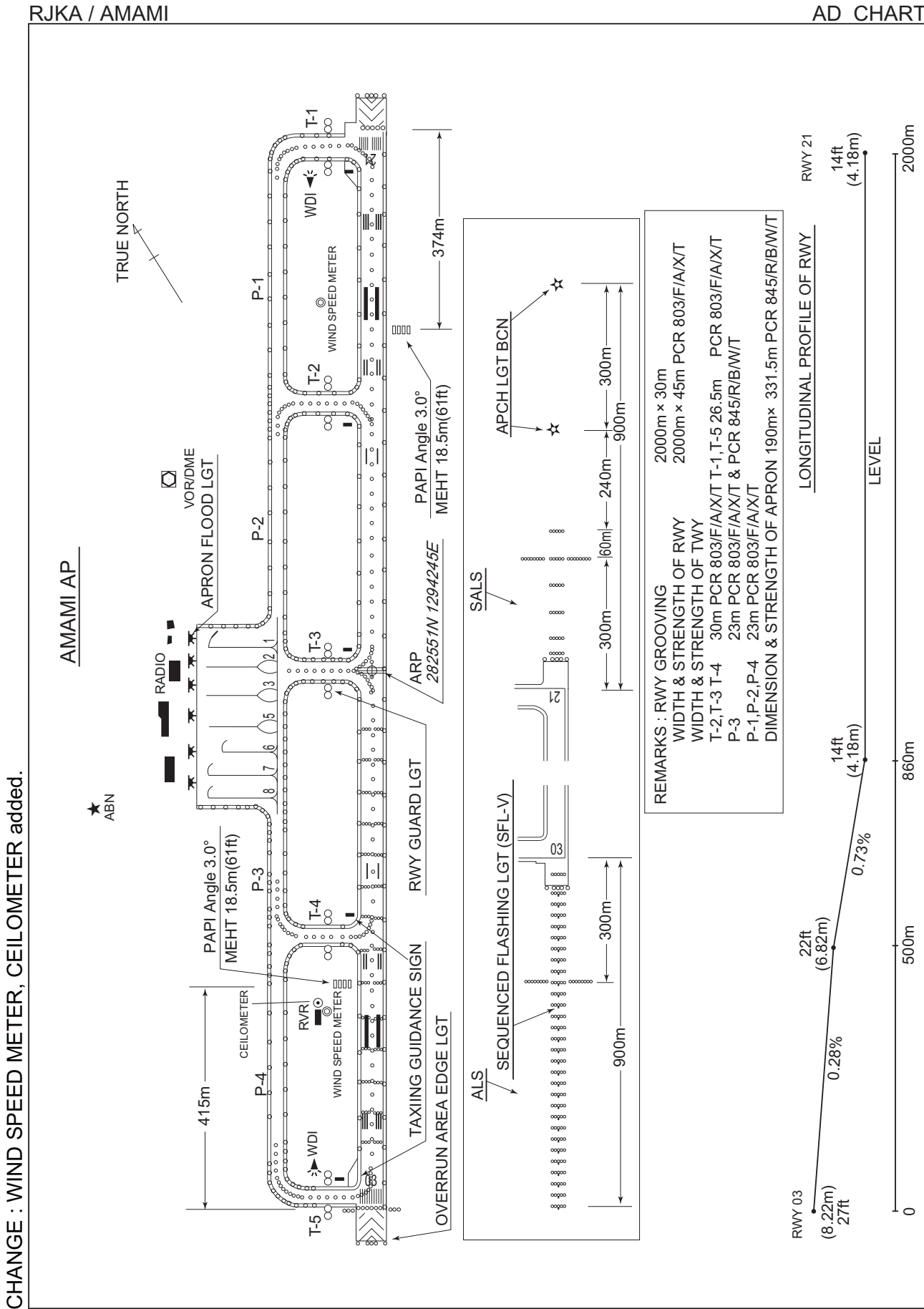




RJKA / AMAMI

RNAV SID

RURIK THREE DEPARTURE

RNP1

Note GNSS required.



RWY21 : 5.0% climb gradient required up to 500FT.

STANDARD DEPARTURE CHART -INSTRUMENT

RJKA / AMAMI

RNAV SID

RURIK THREE DEPARTURE

RWY03

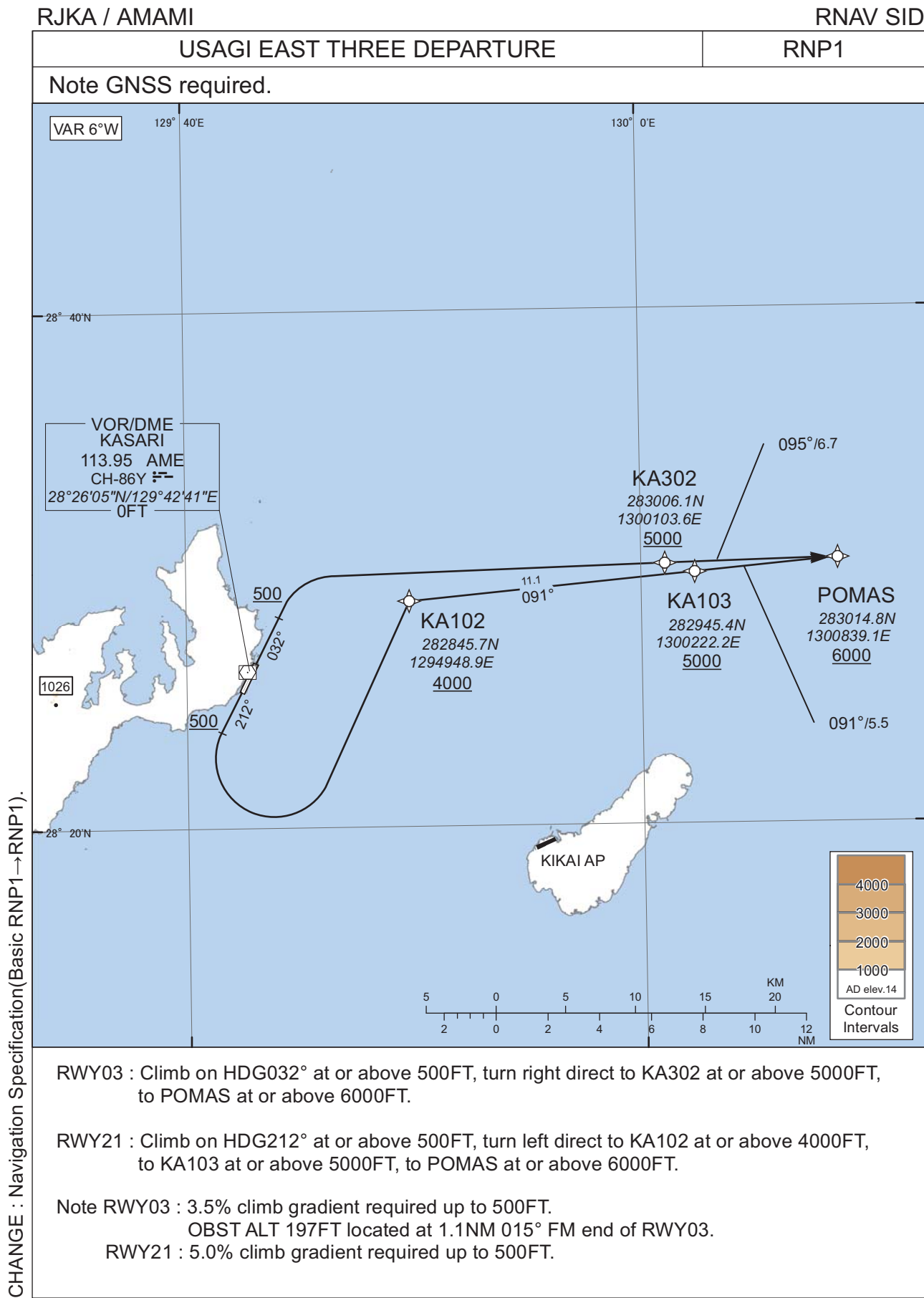
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	032 (025.8)	-6.4	-	-	+500	-	-	RNP1
002	DF	RURIK	-	-	-6.4	-	-	+7000	-	-	RNP1

RWY21

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	212 (205.8)	-6.4	-	-	+500	-	-	RNP1
002	DF	KA102	-	-	-6.4	-	L	+4000	-	-	RNP1
003	TF	RURIK	-	029 (022.2)	-6.4	53.2	-	+7000	-	-	RNP1

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD DEPARTURE CHART -INSTRUMENT



STANDARD DEPARTURE CHART -INSTRUMENT

RJKA / AMAMI

RNAV SID

USAGI EAST THREE DEPARTURE

RWY03

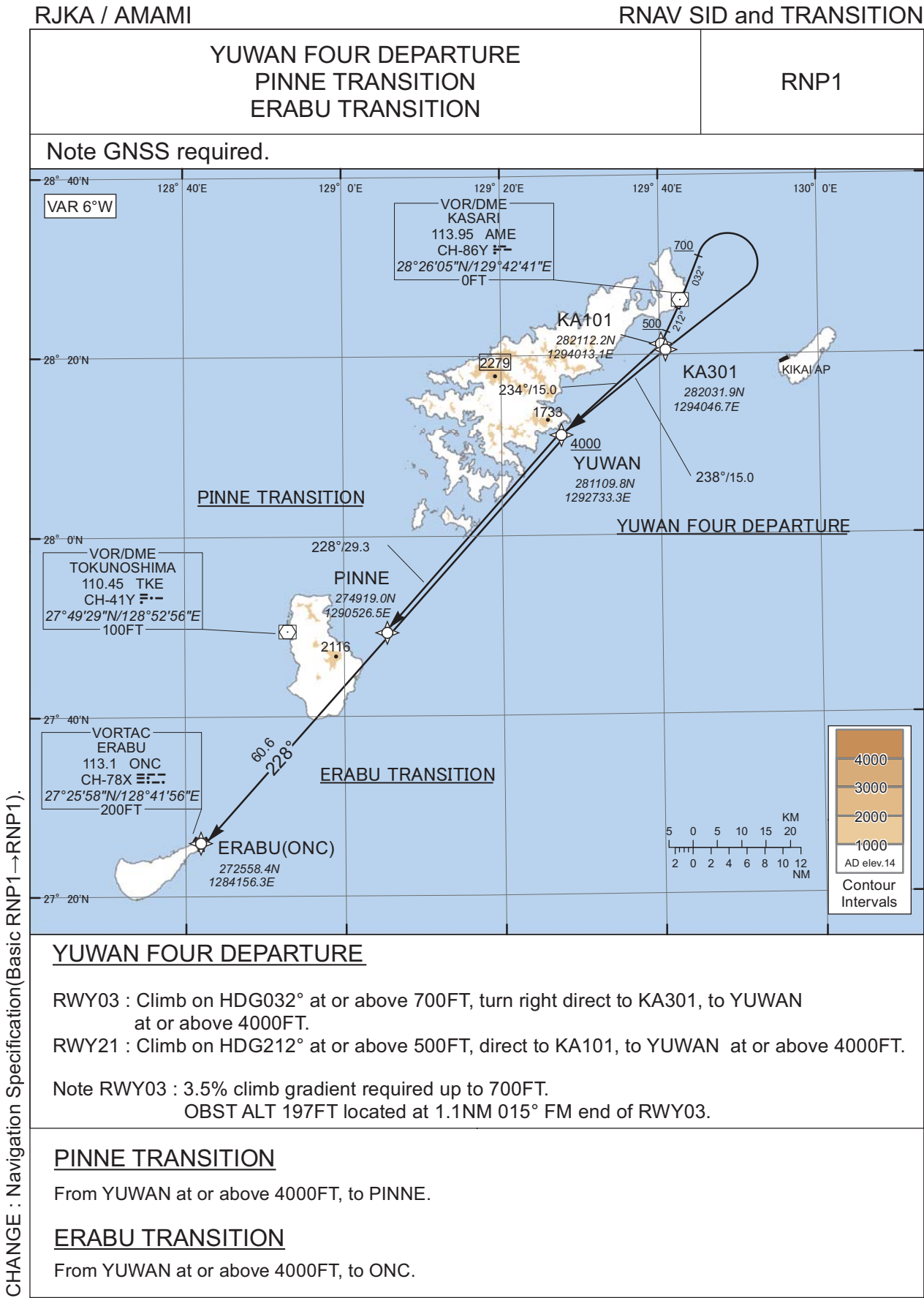
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	032 (025.8)	-6.4	-	-	+500	-	-	RNP1
002	DF	KA302	-	-	-6.4	-	R	+5000	-	-	RNP1
003	TF	POMAS	-	095 (088.7)	-6.4	6.7	-	+6000	-	-	RNP1

RWY21

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	212 (205.8)	-6.4	-	-	+500	-	-	RNP1
002	DF	KA102	-	-	-6.4	-	L	+4000	-	-	RNP1
003	TF	KA103	-	091 (084.8)	-6.4	11.1	-	+5000	-	-	RNP1
004	TF	POMAS	-	091 (084.9)	-6.4	5.5	-	+6000	-	-	RNP1

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD DEPARTURE CHART -INSTRUMENT



STANDARD DEPARTURE CHART -INSTRUMENT

RJKA / AMAMI

RNAV SID and TRANSITION

YUWAN FOUR DEPARTURE

RWY03

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	032 (025.8)	-6.4	-	-	+700	-	-	RNP1
002	DF	KA301	-	-	-6.4	-	R	-	-	-	RNP1
003	TF	YUWAN	-	238 (231.2)	-6.4	15.0	-	+4000	-	-	RNP1

RWY21

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	212 (205.8)	-6.4	-	-	+500	-	-	RNP1
002	DF	KA101	-	-	-6.4	-	-	-	-	-	RNP1
003	TF	YUWAN	-	234 (228.1)	-6.4	15.0	-	+4000	-	-	RNP1

PINNE TRANSITION

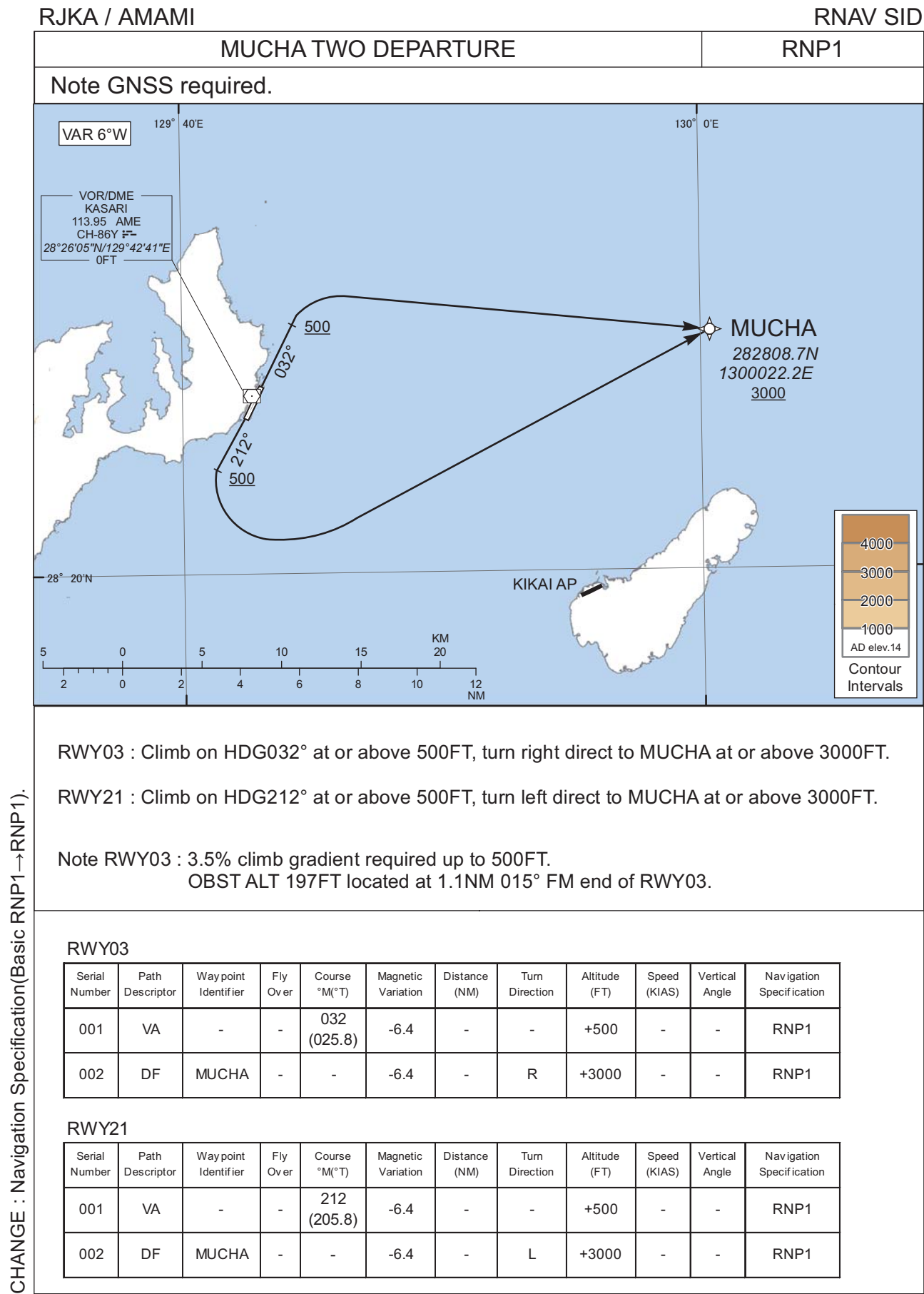
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	YUWAN	-	-	-6.4	-	-	+4000	-	-	RNP1
002	TF	PINNE	-	228 (221.9)	-6.4	29.3	-	-	-	-	RNP1

ERABU TRANSITION

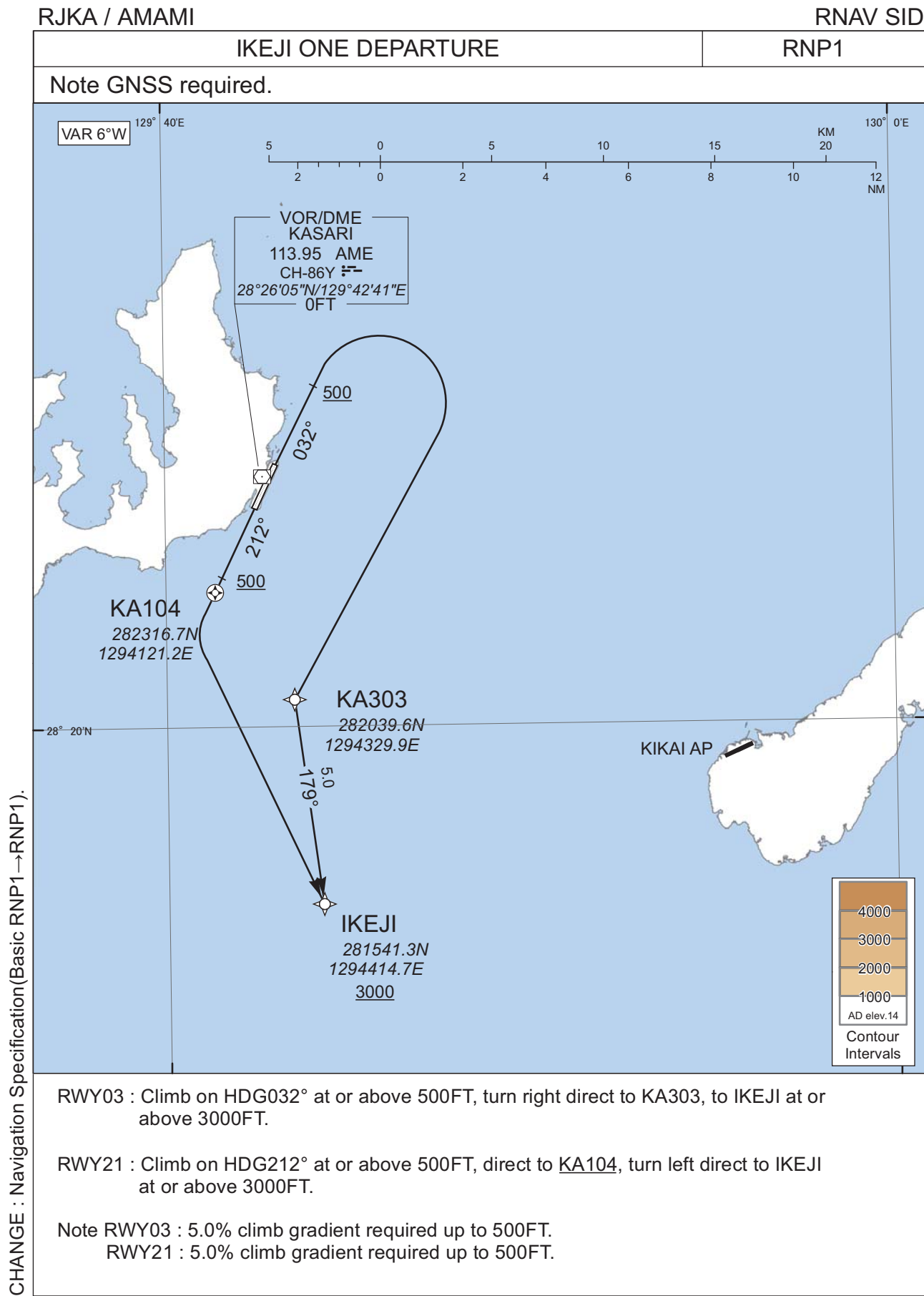
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	YUWAN	-	-	-6.4	-	-	+4000	-	-	RNP1
002	TF	ONC	-	228 (221.9)	-6.4	60.6	-	-	-	-	RNP1

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD DEPARTURE CHART -INSTRUMENT



STANDARD DEPARTURE CHART -INSTRUMENT



STANDARD DEPARTURE CHART -INSTRUMENT

RJKA / AMAMI

RNAV SID

IKEJI ONE DEPARTURE

RWY03

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	032 (025.8)	-6.4	-	-	+500	-	-	RNP1
002	DF	KA303	-	-	-6.4	-	R	-	-	-	RNP1
003	TF	IKEJI	-	179 (172.5)	-6.4	5.0	-	+3000	-	-	RNP1

RWY21

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	212 (205.8)	-6.4	-	-	+500	-	-	RNP1
002	DF	KA104	Y	-	-6.4	-	-	-	-	-	RNP1
003	DF	IKEJI	-	-	-6.4	-	L	+3000	-	-	RNP1

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD DEPARTURE CHART -INSTRUMENT

RJKA / AMAMI

SID

PINNE SIX DEPARTURE

RWY 03 : Climb RWY HDG to 700FT, turn right HDG273° to intercept and proceed...

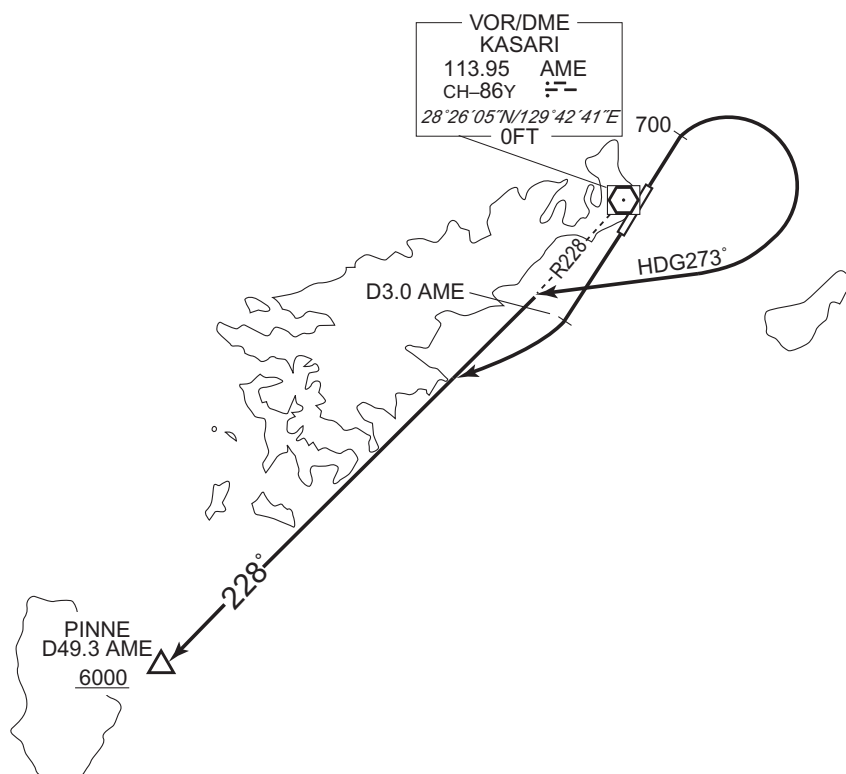
RWY 21 : Climb RWY HDG to AME 3.0DME, turn right,...

... via AME R228 to PINNE.

Cross PINNE at or above 6000FT.

NOTE RWY03 : 3.5% climb gradient required up to 700FT.

OBST ALT 197FT located at 1.1NM 015° FM end of RWY03.



CHANGE : PROC renamed(PINNE SIX DEPARTURE). Note added. ERABU FOUR DEPARTURE abolished.

STANDARD DEPARTURE CHART -INSTRUMENT

RJKA / AMAMI

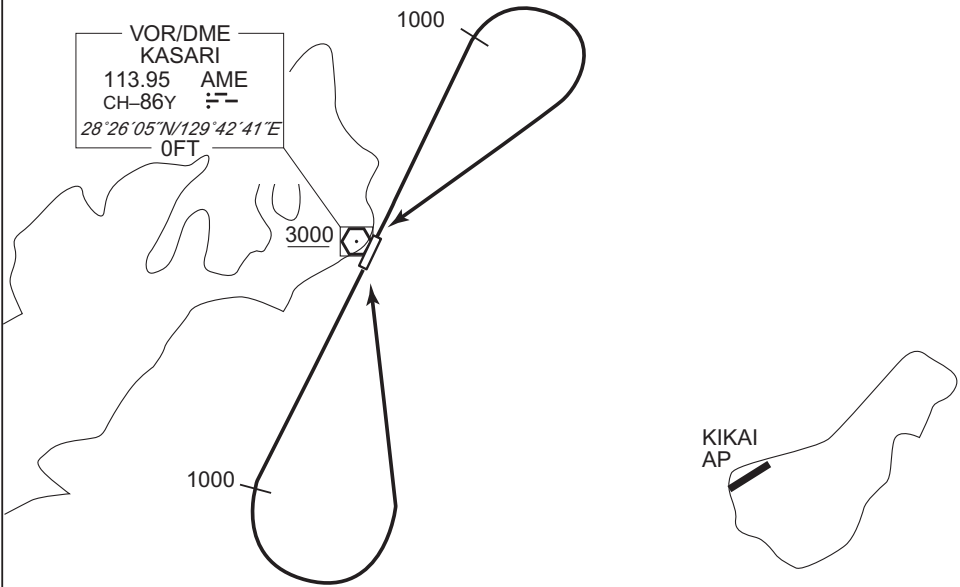
SID

CHANGE : PROC renamed(KASARI REVERSAL THREE DEPARTURE). Note added. POMAS FOUR DEPARTURE abolished.

KASARI REVERSAL THREE DEPARTURE

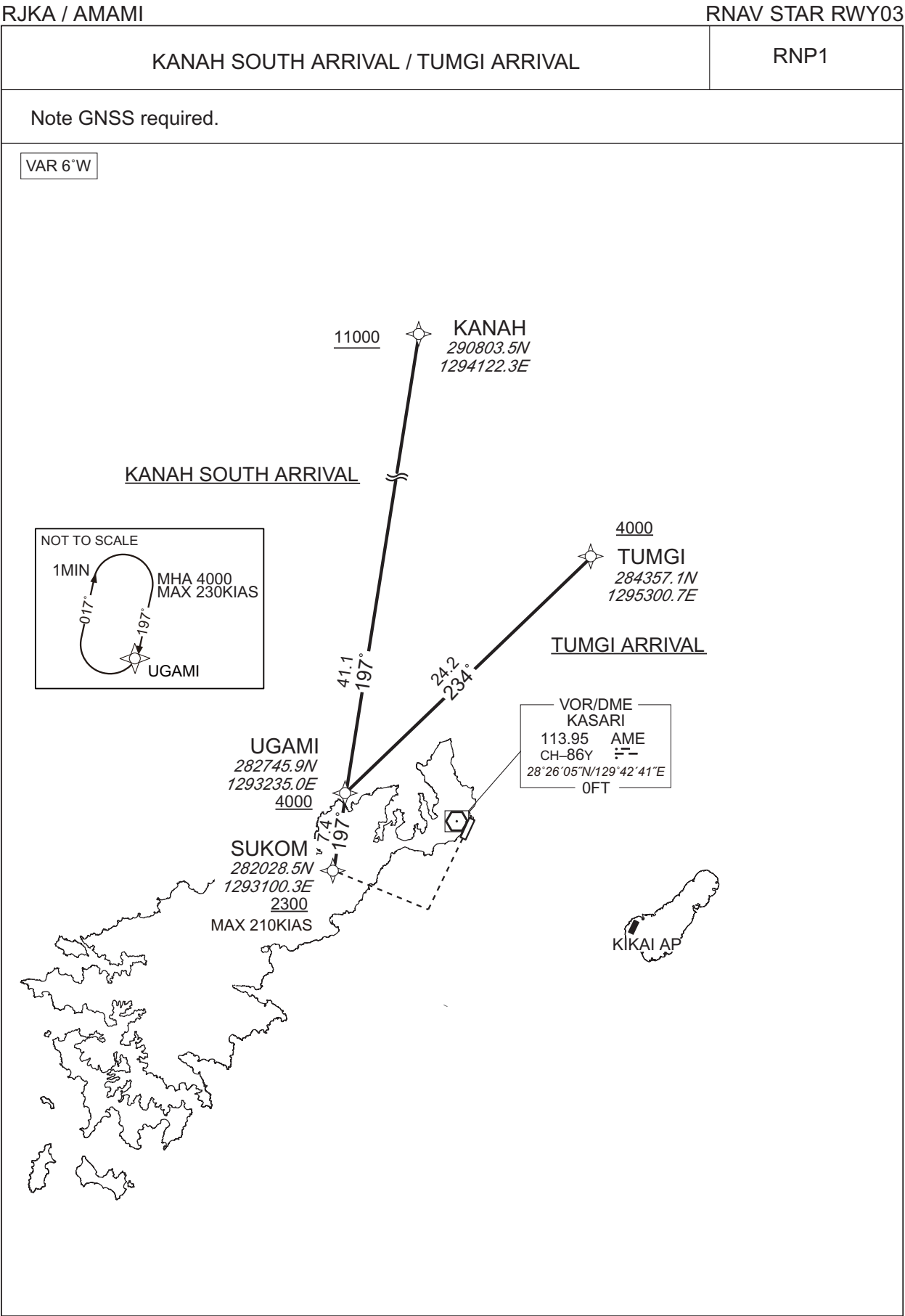
- RWY03 : Climb RWY HDG to 1000FT,turn right,...
- RWY21 : Climb RWY HDG to 1000FT,turn left,...
- ...direct to AME VOR/DME. Cross AME VOR/DME at or above 3000FT.

NOTE RWY03 : 3.5% climb gradient required up to 1000FT.
OBST ALT 197FT located at 1.1NM 015° FM end of RWY03.



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STANDARD ARRIVAL CHART-INSTRUMENT



STANDARD ARRIVAL CHART-INSTRUMENT

RJKA / AMAMI

RNAV STAR RWY03

KANAH SOUTH ARRIVAL

From KANAH at or above 11000FT, to UGAMI at or above 4000FT, to SUKOM at or above 2300FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	KANAH	-	-	-6.3	-	-	+11000	-	-	RNP1
002	TF	UGAMI	-	197 (190.9)	-6.3	41.1	-	+4000	-	-	RNP1
003	TF	SUKOM	-	197 (190.8)	-6.3	7.4	-	+2300	-210	-	RNP1

TUMGI ARRIVAL

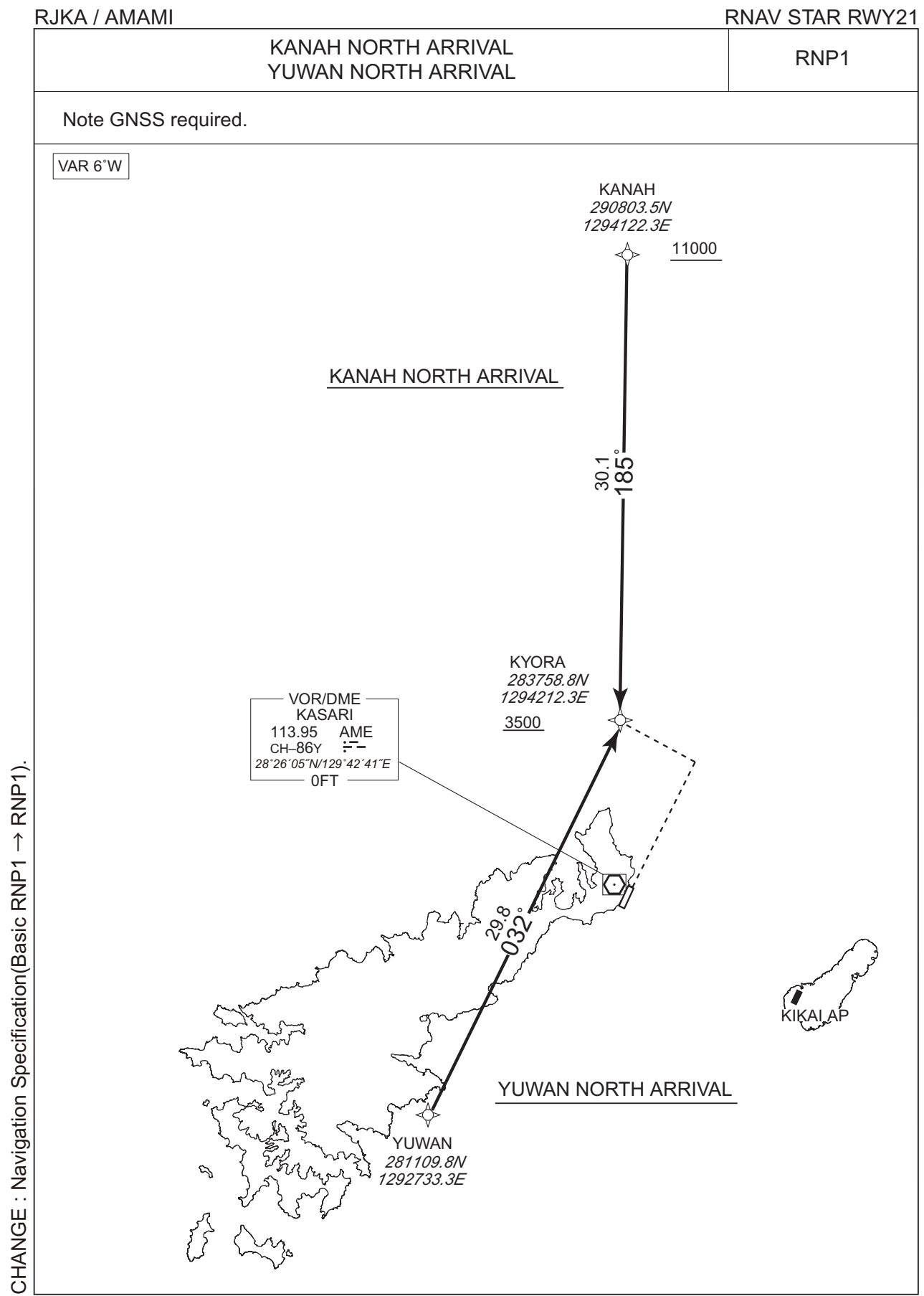
From TUMGI at or above 4000FT, to UGAMI at or above 4000FT, to SUKOM at or above 2300FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	TUMGI	-	-	-6.3	-	-	+4000	-	-	RNP1
002	TF	UGAMI	-	234 (228.0)	-6.3	24.2	-	+4000	-	-	RNP1
003	TF	SUKOM	-	197 (190.8)	-6.3	7.4	-	+2300	-210	-	RNP1

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification
Hold	UGAMI	197 (190.8)	-6.3	1.0 (-14000)	R	4000	FL140	-230 (-14000)	RNP1

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD ARRIVAL CHART-INSTRUMENT



STANDARD ARRIVAL CHART-INSTRUMENT

RJKA / AMAMI

RNAV STAR RWY21

KANAH NORTH ARRIVAL

From KANAH at or above 11000FT, to KYORA at or above 3500FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	KANAH	-	-	-6.2	-	-	+11000	-	-	RNP1
002	TF	KYORA	-	185 (178.6)	-6.2	30.1	-	+3500	-	-	RNP1

YUWAN NORTH ARRIVAL

From YUWAN, to KYORA at or above 3500FT.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	YUWAN	-	-	-6.2	-	-	-	-	-	RNP1
002	TF	KYORA	-	032 (025.6)	-6.2	29.8	-	+3500	-	-	RNP1

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

RJKA / AMAMI

NAHA APP
124.95 – 280.1

ILS – LOC
109.3 IAM
ILS–GP 332.0
ILS–DME CH–30X

AMAMI RADIO
118.15 - 126.2
AFIS provided
by Naha Airport Office

RADAR AVBL

VAR 6°W

MSA 25NM

2900
1700
090°
AME
270°
3400
1900
360°

10NM

VOR/DME
KASARI
113.95 AME
CH-86Y
28°26'05"N/129°42'41"E

• 603

• 601

MAPt(LOC)
D0.8 IAM

LEDUM (FAF)
D4.0 IAM

• 1123

• 1034

• 1592

Turn initiation
within D11.0 AME

MAX Turning speed
200KIAS

LEDUM (FAF) : 282156.17N/1294036.40E

EQPT REQUIRED
DME
VOR

MHA 3000
MAX 230KIAS

NOT TO
SCALE

036°
216°
AME
VOR/DME

NM to IAM

FAF

3

2

1

MAPt

ALT (3.0° APCH Path)

1287

977

658

339

–

AME 3000

Turn initiation within D11.0 AME

1800

032°

227°

LEDUM (FAF)

1300

1287(LOC)

GP 3.0°

032°

VDP (LOC)

MAPt (LOC)

700 (686)

MDA

RDH 54

4.0

0.8

0.8

0.2

DME to IAM

3.8

0.7

0.6

0

NM to THR

MISSED APPROACH

Climb on HDG032° to 3000FT,
turn right, direct to AME
VOR/DME and hold.
Contact NAHA APP.

Timing not authorized for defining the MAPt.

MINIMA

THR elev. 27

AD elev. 14

CAT

CAT I

LOC

CIRCLING

DA(H)

RVR/
CMV

MDA(H)

RVR/
CMV

MDA(H)

VIS

A

B

C

D

227 (200)

550

280 (266)

800

500 (486)

1600

2400

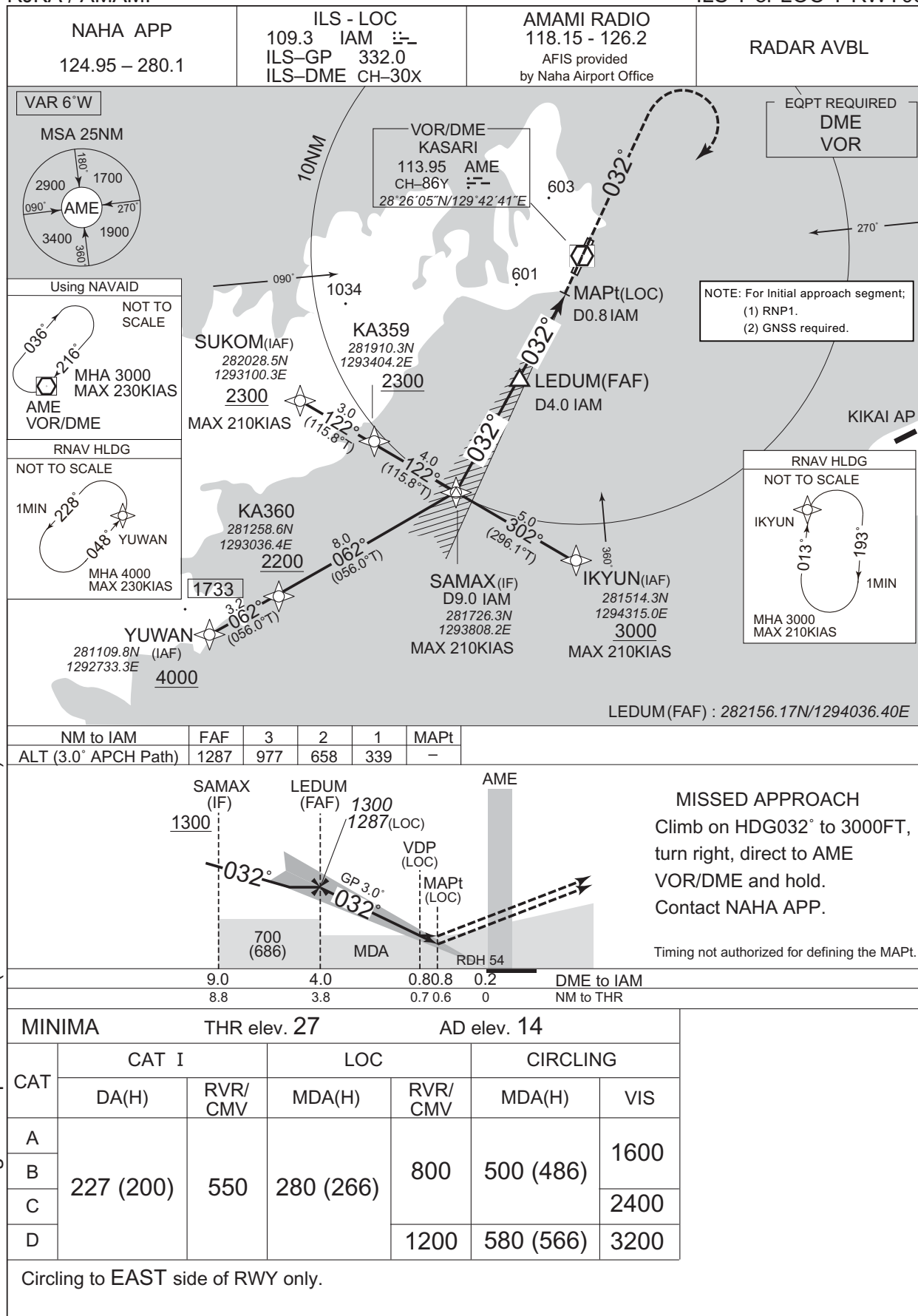
3200

Circling to EAST side of RWY only.

INSTRUMENT APPROACH CHART

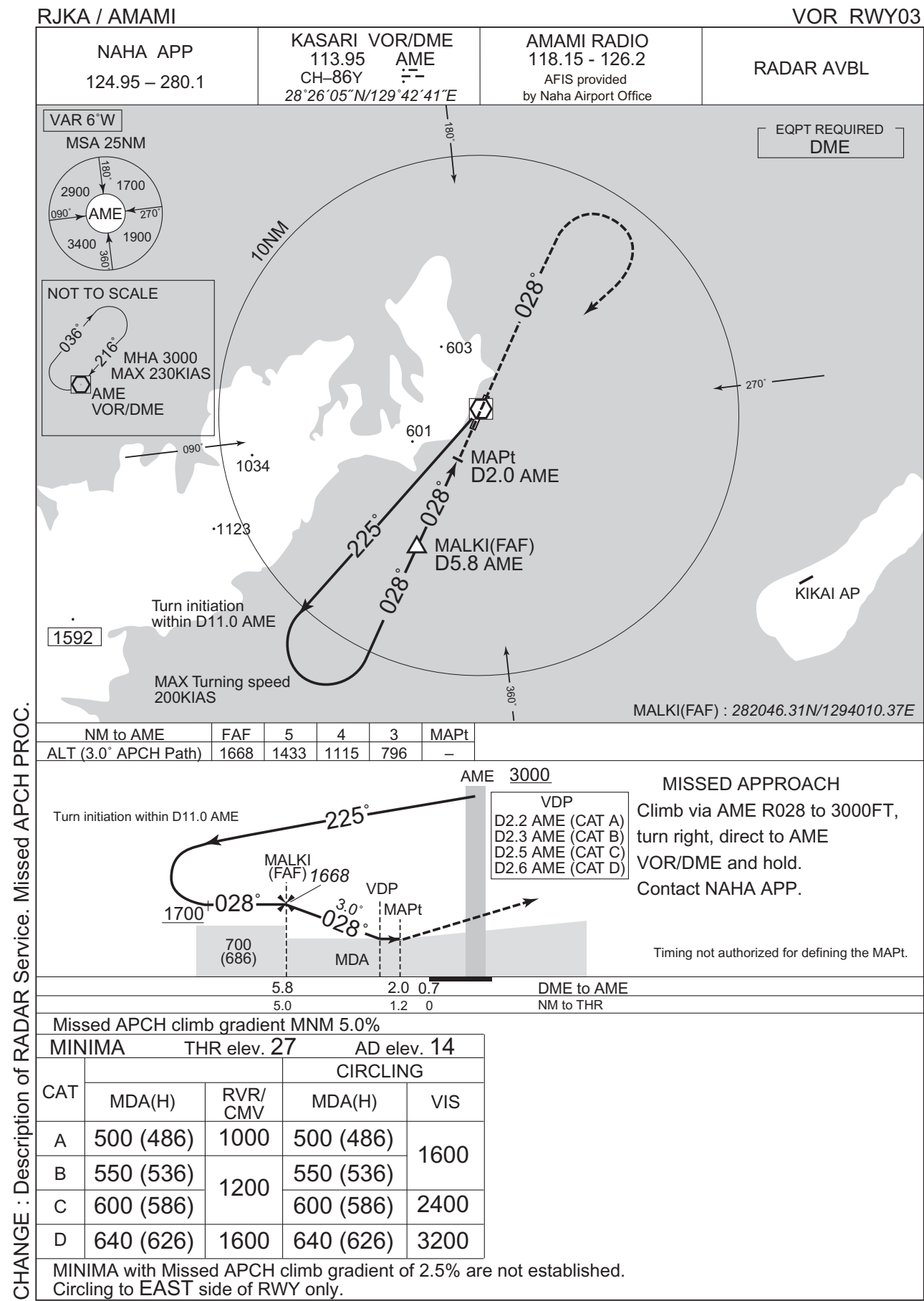
RJKA / AMAMI

ILS Y or LOC Y RWY03



CHANGE : Navigation Specification(Basic RNP1 → RNP1).

INSTRUMENT APPROACH CHART

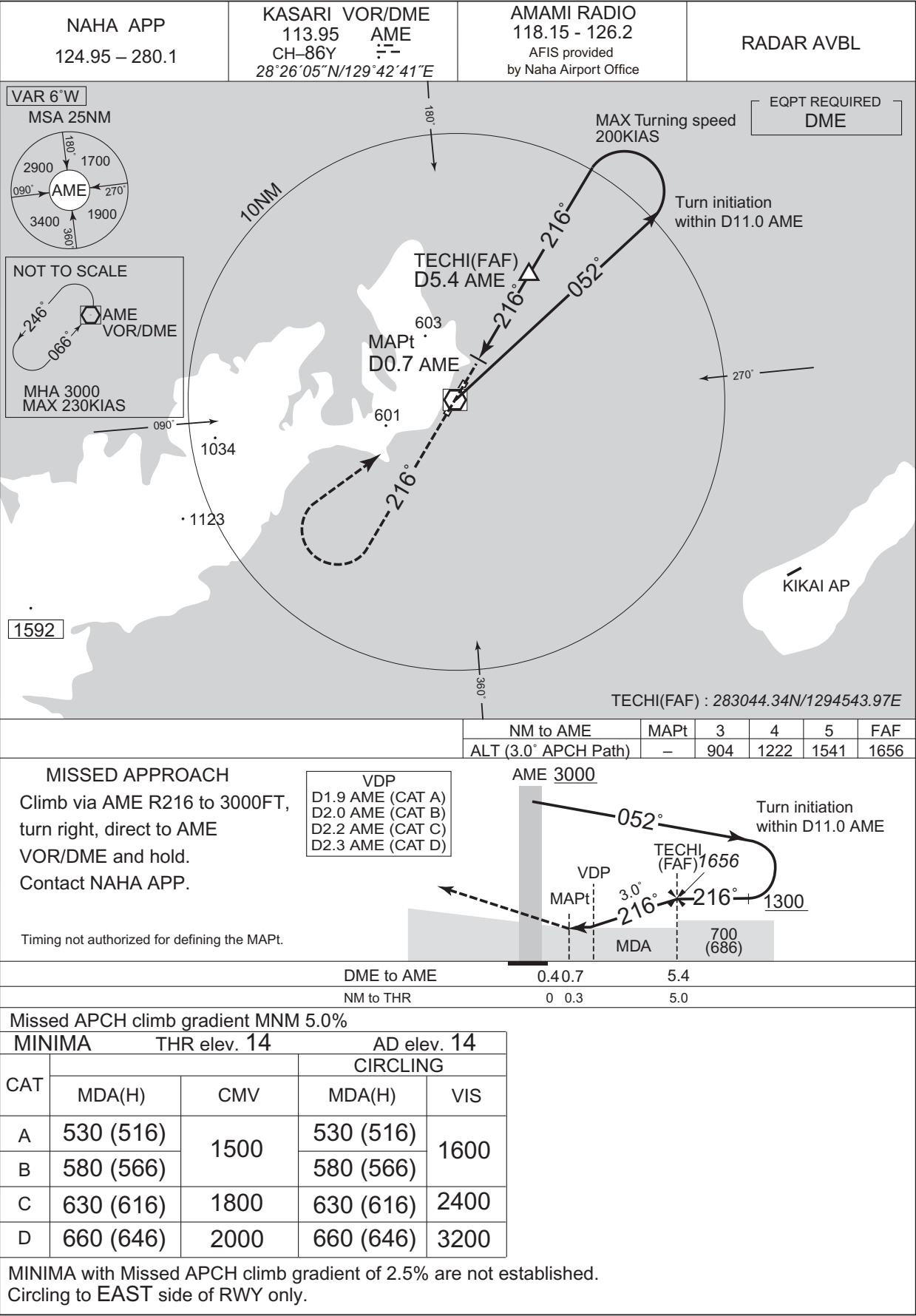


CHANGE : Description of RADAR Service. Missed APCH PROC.

INSTRUMENT APPROACH CHART

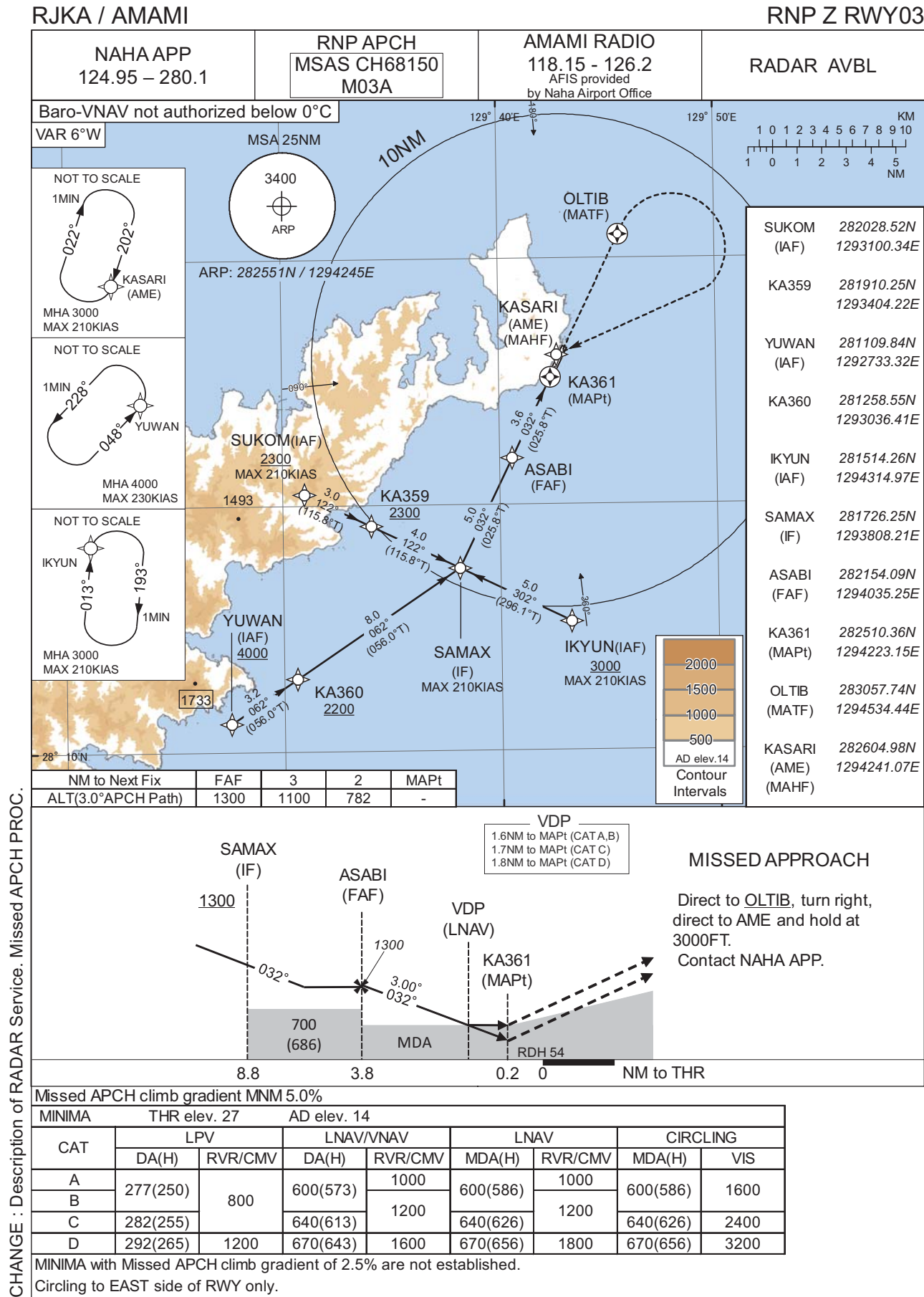
RJKA / AMAMI

VOR RWY21



CHANGE : Description of RADAR Service. Missed APCH PROC.

INSTRUMENT APPROACH CHART



INSTRUMENT APPROACH CHART

RJKA / AMAMI

RNP Z RWY03

FAS DATA BLOCK

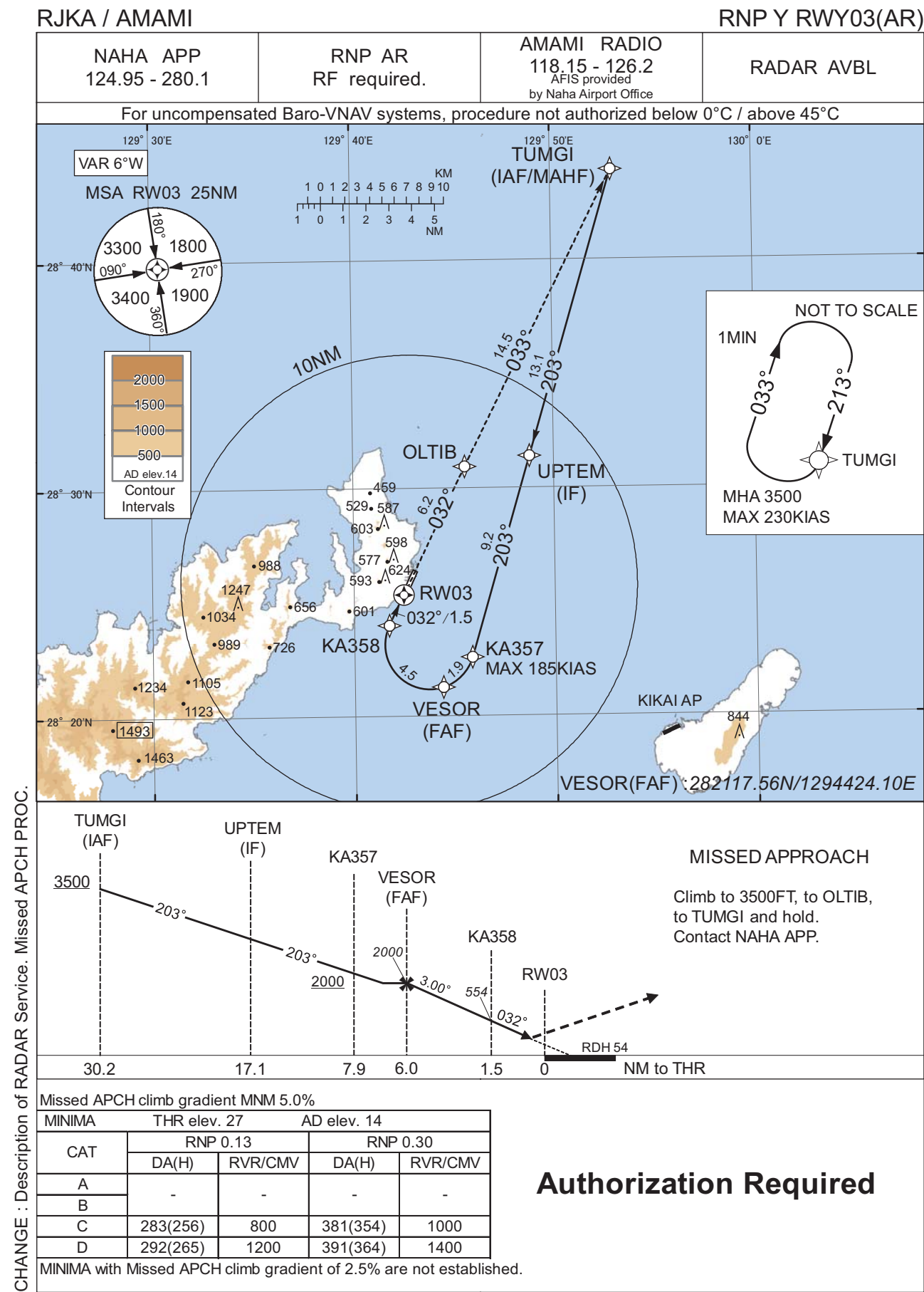
Operation type	0	LTP/FTP ellipsoidal height	+00373
SBAS service provider identifier	2	FPAP latitude	282619.5815N
Airport identifier	RJKA	FPAP longitude	1294301.2745E
Runway	03	Threshold crossing height	00016.5
Approach performance designator	0	TCH units selector	1
Route indicator	Z	Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M03A	∟ length offset	0000
LTP/FTP latitude	282521.1620N	HAL	40.0
LTP/FTP longitude	1294229.1275E	VAL	50.0
CRC remainder	2C1731A7		

Required additional data

LTP/FTP orthometric height	8.5
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CHANGE : New PROC.

INSTRUMENT APPROACH CHART



INSTRUMENT APPROACH CHART

RJKA / AMAMI

RNP Y RWY03(AR)

Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	TUMGI	-	-	-6.4	-	-	+3500	-	-	-
002	TF	UPTM	-	203 (196.4)	-6.4	13.1	-	-	-	-	0.3
003	TF	KA357	-	203 (196.4)	-6.4	9.2	-	+2000	-185	-	0.3
004	RF Center: KARF2 r=1.95NM	VESOR	-	-	-6.4	1.9	R	2000	-	-	0.3
005	RF Center: KARF2 r=1.95NM	KA358	-	-	-6.4	4.5	R	554	-	-3.00	0.13 0.30
006	TF	RW03	Y	032 (025.8)	-6.4	1.5	-	81	-	-3.00/54	0.13 0.30
007	TF	OLTIB	-	032 (025.8)	-6.4	6.2	-	-	-	-	1.0
008	TF	TUMGI	-	033 (026.7)	-6.4	14.5	-	3500	-	-	1.0

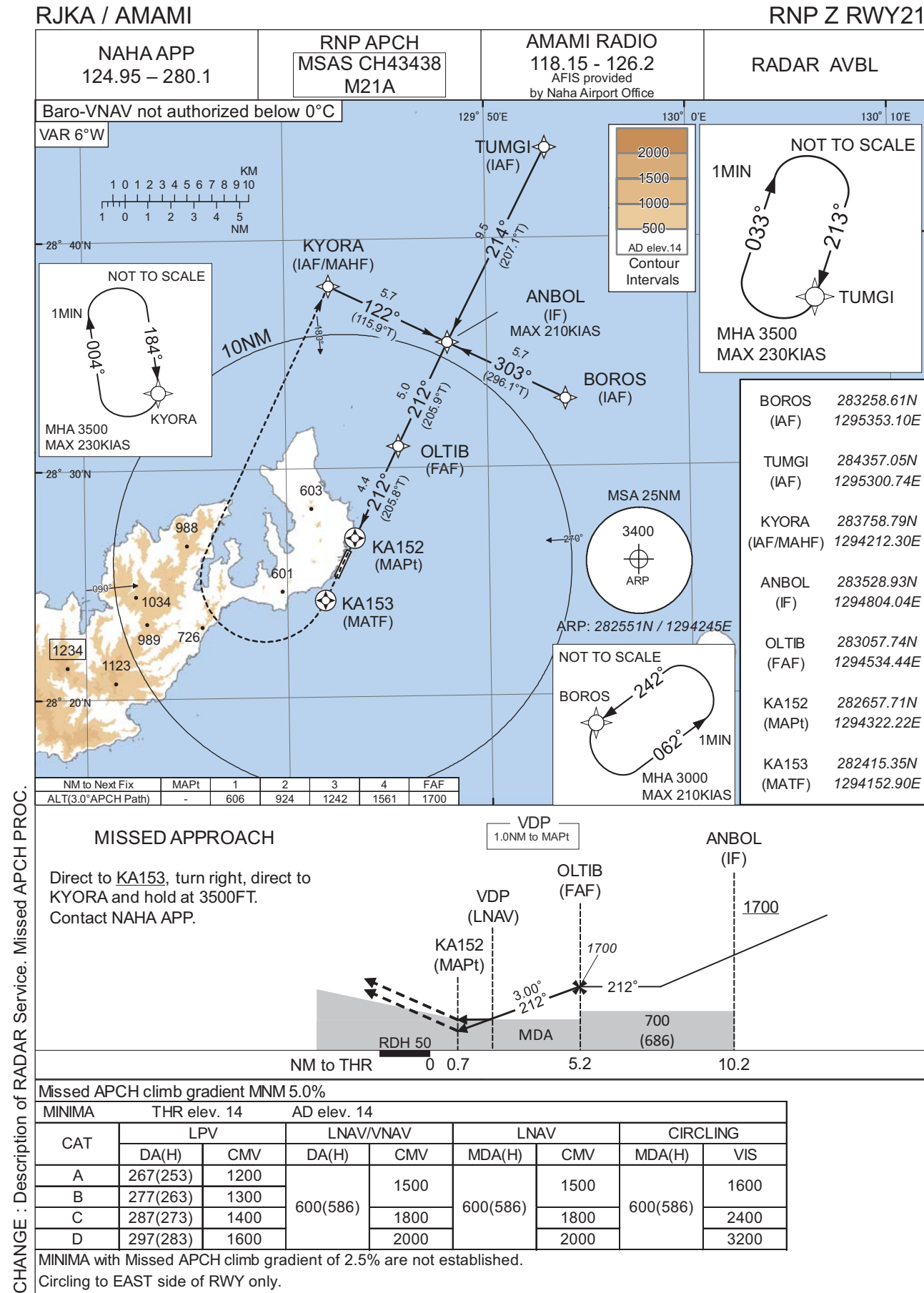
Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	TUMGI	213 (207.0)	-6.3	1.0 (-14000)	R	3500	FL140	-230 (-14000)	1.0

Waypoint Coordinates

Waypoint Identifier	Coordinates	RF Arc Center Identifier	Coordinates
TUMGI	284357.05N / 1295300.74E	KARF2	282309.49N / 1294344.25E
UPTM	283124.97N / 1294848.31E		
KA357	282236.27N / 1294551.51E		
VESOR	282117.56N / 1294424.10E		
KA358	282400.82N / 1294144.91E		
RW03	282521.18N / 1294229.11E		
OLTIB	283057.74N / 1294534.44E		

CHANGE : PROC renamed. PROC course. UPTM, KA358, OLTIB established. YANGO, KA353, JOUGO, KA351, HABUH abolished.
RNP Value FM UPTM to VESOR. RF Arc Center Identifier (KARF2 added, KARF1 deleted). Waypoint Coordinates (RW03). VAR.

INSTRUMENT APPROACH CHART



INSTRUMENT APPROACH CHART

RJKA / AMAMI

RNP Z RWY21

FAS DATA BLOCK

Operation type	0	LTP/FTP ellipsoidal height	+00333
SBAS service provider identifier	2	FPAP latitude	282521.1620N
Airport identifier	RJKA	FPAP longitude	1294229.1275E
Runway	21	Threshold crossing height	00015.0
Approach performance designator	0	TCH units selector	1
Route indicator	Z	Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M21A	∟ length offset	0000
LTP/FTP latitude	282619.5815N	HAL	40.0
LTP/FTP longitude	1294301.2745E	VAL	50.0
CRC remainder	A4D8FDB0		

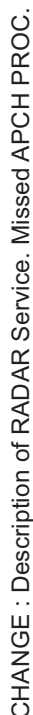
Required additional data

LTP/FTP orthometric height	4.5
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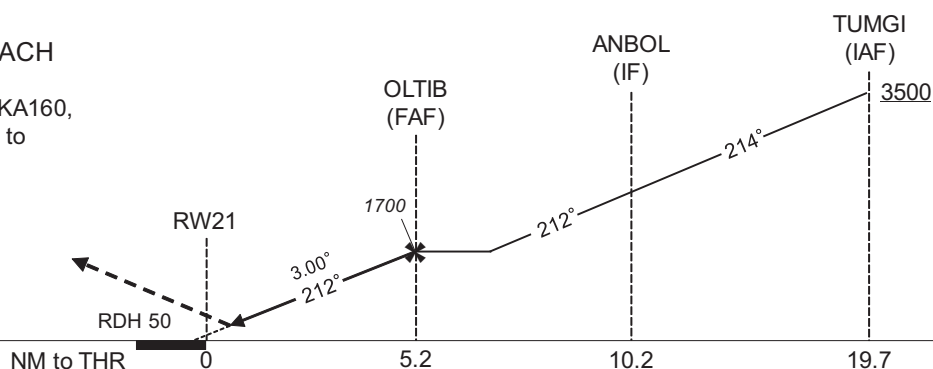
CHANGE : FAS DATA BLOCK and Required additional data established.

RJKA / AMAMI

RNP Y RWY21(AR)



Climb to 3500FT, to KA160,
to KA161, to KA162, to
TUMGI and hold.
Contact NAHA APP.



Missed APCH climb gradient MNM 5.0%

MINIMA		THR elev. 14		AD elev. 14	
CAT	RNP 0.14		RNP 0.30		
	DA(H)	CMV	DA(H)	CMV	
A	-	-	-	-	
B					
C	287(273)	1400	365(351)	1600	
D	297(283)	1600	387(373)	1800	

MINIMA with Missed APCH climb gradient of 2.5% are not established.

Authorization Required

INSTRUMENT APPROACH CHART

RJKA / AMAMI

RNP Y RWY21(AR)

CHANGE : PROC course. ANBOL, OLTIB, KA160 established. IMORE, HABUH, KA101 abolished. Waypoint Coordinates (RW21). VAR.

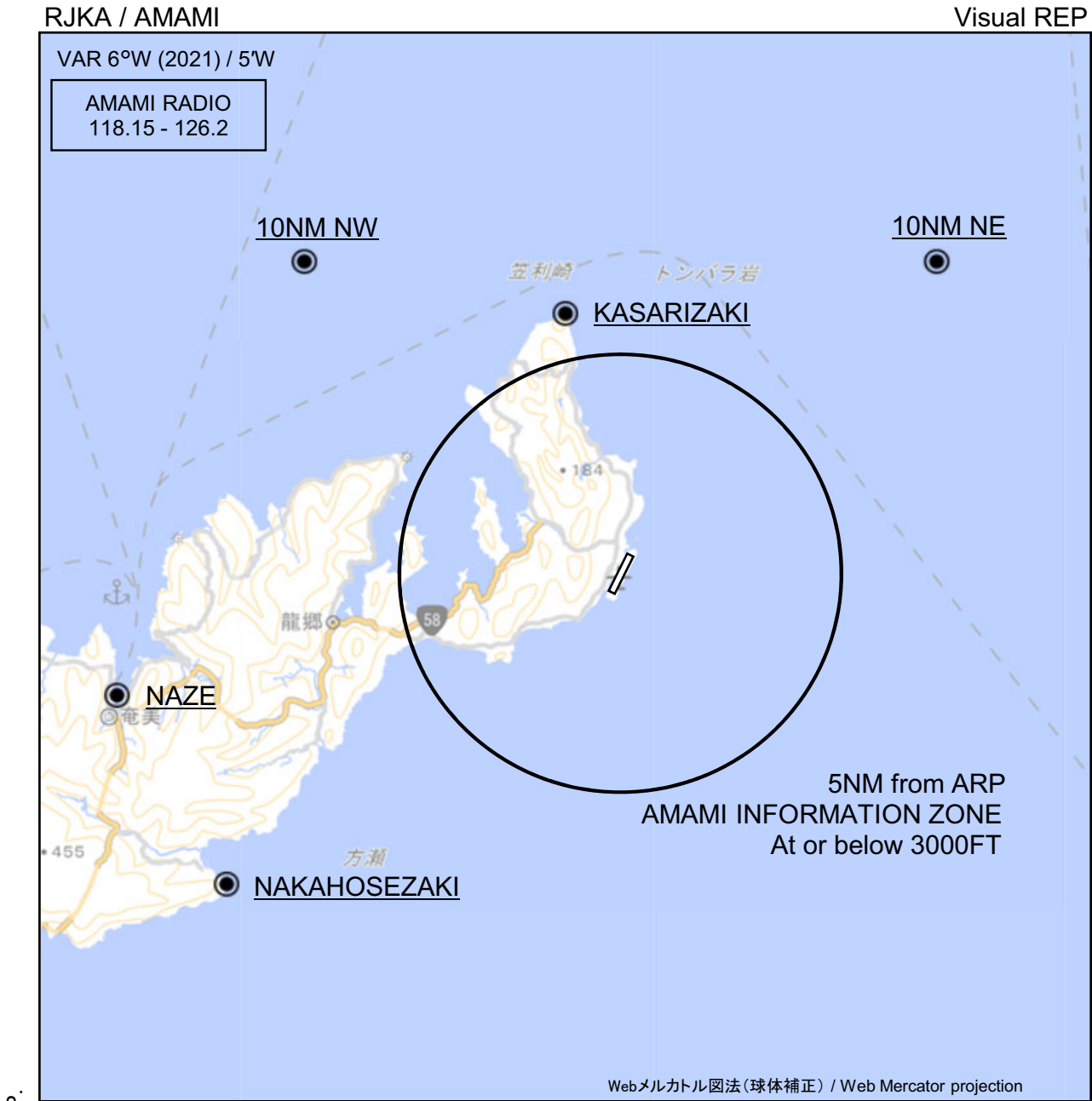
Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	TUMGI	-	-	-6.4	-	-	+3500	-	-	-
002	TF	ANBOL	-	214 (207.1)	-6.4	9.5	-	-	-	-	1.0
003	TF	OLTIB	-	212 (205.9)	-6.4	5.0	-	1700	-	-	1.0
004	TF	RW21	Y	212 (205.8)	-6.4	5.2	-	64	-	-3.00/50	0.14 0.30
005	TF	KA160	-	212 (205.8)	-6.4	5.7	-	-	-	-	1.0
006	TF	KA161	-	302 (296.0)	-6.4	11.0	-	-	-	-	1.0
007	TF	KA162	-	017 (010.7)	-6.4	10.0	-	-	-	-	1.0
008	TF	TUMGI	-	074 (067.1)	-6.4	20.9	-	3500	-	-	1.0

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	TUMGI	213 (207.0)	-6.3	1.0 (-14000)	R	3500	FL140	-230 (-14000)	1.0

Waypoint Coordinates

Waypoint Identifier	Coordinates
TUMGI	284357.05N / 1295300.74E
ANBOL	283528.93N / 1294804.04E
OLTIB	283057.74N / 1294534.44E
RW21	282619.61N / 1294301.26E
KA160	282111.47N / 1294011.84E
KA161	282600.04N / 1292858.96E
KA162	283551.31N / 1293105.90E



CHANGE : Map updated. BRG/DIST from ARP.

※図中に標高を示す数字がある場合、単位はメートル(m)である。The unit of measurement used to express elevation is meter(m).

Call sign	BRG / DIST from ARP	Remarks
10NM NE	045°T / 10.0NM	海上 Over the sea
10NM NW	315°T / 10.0NM	海上 Over the sea
笠利崎 Kasarizaki	348°T / 6.0NM	灯台 Lighthouse
名瀬 Naze	256°T / 11.7NM	港 Harbor
仲干瀬崎 Nakahosezaki	231°T / 11.4NM	岬 Cape

RJKA / AMAMI

Minimum Vectoring Altitude CHART

